European Journal of Cancer xxx (xxxx) xxx

Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.ejcancer.com

Original Research

# Apalutamide plus androgen deprivation therapy in clinical subgroups of patients with metastatic castration-sensitive prostate cancer: A subgroup analysis of the randomised clinical TITAN study

Axel S. Merseburger [a,*], Neeraj Agarwal [b], Amitabha Bhaumik [c], Florence Lefresne [d], Laurence I. Karsh [e], Andrea J. Pereira de Santana Gomes [f], Álvaro Juárez Soto [g], Robert W. Given [h], Sabine D. Brookman-May [d,i], Suneel D. Mundle [j], Sharon A. McCarthy [j], Hirotsugu Uemura [k], Simon Chowdhury [l,m], Kim N. Chi [n], Anders Bjartell [o]

[a] *University Hospital Schleswig-Holstein, Campus Lübeck, Ratzeburger Allee 160, 23538 Lübeck, Germany*
[b] *Huntsman Cancer Institute, University of Utah, 2000 Circle of Hope Drive, Suite 5726, Salt Lake City, UT 84112, USA*
[c] *Janssen Research & Development, 1125 Trenton-Harbourton Road, Titusville, NJ 08560, USA*
[d] *Janssen Research & Development, 1400 McKean Road, Spring House, PA 19477, USA*
[e] *The Urology Center of Colorado, 2777 Mile High Stadium Circle, Denver, CO 80211, USA*
[f] *Liga Norte Riograndense Contra O Cancer, Av Miguel Castro, 1355, Dix-Sept Rosado, Natal 59075–740, Brazil*
[g] *Hospital Universitario de Jerez de la Frontera, Ronda de Circunvalación sln, 11407 Jerez de la Frontera, Cádiz, Spain*
[h] *Urology of Virginia, Eastern Virginia Medical School, 825 Fairfax Ave., Suite 310, Norfolk, VA 23507, USA*
[i] *Ludwig-Maximilians-University, Geschwister-Scholl-Platz 1, D-80539 München, Germany*
[j] *Janssen Research & Development, 700 US Highway 202 S, Raritan, NJ 08869, USA*
[k] *Kindai University Faculty of Medicine, 377–2 Ohno-Higashi, Osakasayama, Osaka 589–8511, Japan*
[l] *Guy's, King's, and St Thomas' Hospitals, Great Maze Pond, London SE1 9RT, UK*
[m] *Sarah Cannon Research Institute, 93 Harley St, Marylebone, London W1G 6AD, UK*
[n] *BC Cancer and Vancouver Prostate Centre, 600 West 10th Avenue, Vancouver V5Z 1L3, British Columbia, Canada*
[o] *Skåne University Hospital, Lund University, Jan Waldenströms gata 5, plan 2, 20502 Malmö, Sweden*

Received 30 March 2023; Received in revised form 31 July 2023; Accepted 6 August 2023
Available online xxxx

* *Correspondence to*: Department of Urology, University Hospital Schleswig-Holstein, Campus Lübeck, Ratzeburger Allee 160, 23538 Lübeck, Germany.

*E-mail addresses:* axel.merseburger@uksh.de (A.S. Merseburger), neeraj.agarwal@hci.utah.edu (N. Agarwal), abhaumik@its.jnj.com (A. Bhaumik), flefresn@its.jnj.com (F. Lefresne), larrykarsh@gmail.com (L.I. Karsh), pesquisaclinica.juliana@liga.org.br (A.J. Pereira de Santana Gomes), alvaro.juarez01@gmail.com (Á.J. Soto), rgiven@urologyofva.net (R.W. Given), sbrookma@its.jnj.com (S.D. Brookman-May), smundle@its.jnj.com (S.D. Mundle), smccar15@its.jnj.com (S.A. McCarthy), huemura@med.kindai.ac.jp (H. Uemura), simonchowdhuryuk@yahoo.co.uk (S. Chowdhury), kchi@bccancer.bc.ca (K.N. Chi), anders.bjartell@med.lu.se (A. Bjartell).

https://doi.org/10.1016/j.ejca.2023.113290
0959-8049/© 2023 The Authors. Published by Elsevier Ltd.

UNCORRECTED PROOF

Please cite this article as: A.S. Merseburger et al., Apalutamide plus androgen deprivation therapy in clinical subgroups of patients with metastatic castration-sensitive prostate cancer: A subgroup analysis of the randomised clinical TITAN study, European Journal of Cancer, https://doi.org/10.1016/j.ejca.2023.113290

**KEYWORDS**
High-volume;
Low-volume;
Synchronous;
Metachronous;
Oligometastatic;
Polymetastatic

**Abstract** ***Background:*** Whetherdisease burden in patients with metastatic castration-sensitive prostate cancer (mCSPC) predicts treatment outcomes is unknown. We assessed apalutamide treatment effect in TITAN patients with mCSPC by disease volume, metastasis number and timing of metastasis presentation.

***Methods:*** These protocol-defined and *post hoc* analyses of the phase III randomised TITAN study evaluated clinical outcomes in patients receiving 240 mg/day apalutamide ($n = 525$) or placebo ($n = 527$) plus androgen-deprivation therapy (ADT). Subgroups were defined by volume (high: visceral and ≥1 bone metastases or ≥4 bone lesions with ≥1 beyond vertebral column/pelvis), development of metastases per conventional imaging (synchronous: at initial diagnosis; metachronous: after localised disease) and oligometastases (≤5 bone-only metastases) or polymetastases (> 5 in bone ± other locations or ≤5 in bone plus other locations). Overall survival (OS), radiographic or second progression-free survival, and time to prostate-specific antigen progression or castration resistance were assessed using Cox proportional hazards models.

***Results:*** Of 1052 patients, 63%, 81%, 54%, 27%, 5.7%, and 8.0% had high-volume, synchronous, synchronous/high-volume, synchronous/low-volume, metachronous/high-volume, and metachronous/low-volume disease, respectively. The OS benefit favoured apalutamide plus ADT versus ADT alone in synchronous/high-volume (HR = 0.68 [95% confidence interval: 0.53–0.87]; $p = 0.002$), synchronous/low-volume (0.65 [0.40–1.05]; $p = 0.08$), metachronous/high-volume (0.69 [0.33–1.44]; $p = 0.32$) and metachronous/low-volume (0.22 [0.09–0.55]; $p = 0.001$) subgroups. Apalutamide improved other clinical outcomes regardless of subgroup, with similar safety profiles. Most favourable outcomes were observed in oligometastatic disease.

***Conclusion:*** TITAN patients derived a robust benefit with apalutamide plus ADT regardless of disease volume and timing of metastasis presentation without differences in safety, supporting early apalutamide intensification in mCSPC.

***Clinical Trial Registration:*** NCT02489318.

© 2023 The Authors. Published by Elsevier Ltd.

UNCORRECTED PROOF

## 1. Introduction

Metastatic burden is prognostic for patients with metastatic castration-sensitive prostate cancer (mCSPC). High-volume presentation with metastases at initial diagnosis (synchronous disease) and low-volume presentation with metastases occurring after primary treatment for localised prostate cancer (metachronous disease) have been associated with less and more favourable prognosis, respectively [1]. Oligometastatic versus polymetastatic disease and specific anatomical locations of metastases may also be important prognostic variables, although further validation of their prognostic value is needed [1–3].

Effective approaches for patients with mCSPC and various disease characteristics are being investigated. Thus, treatment intensification with abiraterone acetate plus prednisone (AAP) added to standard of care (androgen-deprivation therapy [ADT] ± docetaxel), with or without radiation therapy, in PEACE-1 improved radiographic progression-free survival (rPFS) in both high- and low-volume subgroups and overall survival (OS) in the high-volume subgroup [4]. The Advanced Prostate Cancer Consensus Conference concurred on the importance of distinguishing synchronous from metachronous low-volume mCSPC and strongly agreed on treating both groups with ADT plus an androgen receptor (AR) pathway inhibitor and adding primary radiation therapy for synchronous/low-volume mCSPC [5]. Long follow-up analysis of PEACE-1 showed that the addition of prostate radiotherapy to systemic therapy with AAP plus ADT, with or without docetaxel, for synchronous/low-volume mCSPC did not improve OS but improved rPFS, time to castration resistance and rates of serious genitourinary events [6]. Treatment intensification with enzalutamide and ADT in ARCHES improved outcomes in mCSPC irrespective of volume, risk, metastatic burden or metachronous/synchronous disease presentation [7–13]. Addition of docetaxel to ADT increased OS in high-volume but not low-volume mCSPC in CHAARTED and GETUG-AFU15 studies [14–17]. Meta-analysis of these two studies and STAMPEDE showed that doublet treatment improved progression-free survival and OS in all patients, except the metachronous/low-volume subgroup, but was best for patients with high-volume disease [18,19]. Triplet treatment with darolutamide, ADT, and docetaxel in ARASENS improved OS in high-volume and synchronous/high-volume disease [20]. Enzalutamide plus ADT showed survival benefits in patients with metachronous/low-volume disease compared with nonsteroidal antiandrogens plus ADT in ENZAMET [3]. Enzalutamide plus ADT combination was also more beneficial for ENZAMET patients

with low-volume disease who did not receive docetaxel compared with those who did or those with high-volume disease [21].

In TITAN (ClinicalTrials.com: NCT02489318), a phase III, randomised, double-blind placebo-controlled study, apalutamide plus ADT significantly reduced the risk of rPFS and delayed the time to second progression-free survival (PFS2), prostate-specific antigen (PSA) progression and the onset of castration resistance compared with ADT alone [22,23]. We have reported the treatment benefit of apalutamide in a broad population of patients with mCSPC and with synchronous/metachronous or high-/low-volume disease, or with different metastatic burdens using early data cutoffs [22–26]. However, whether there is a differential benefit from apalutamide plus ADT between patient subgroups at the final cutoff warrants further study.

Herein, we report an updated and detailed assessment of the treatment effect and safety profile of apalutamide within specific TITAN subgroups characterised by disease volume, number of metastases and synchronous/metachronous metastasis presentation using final data cutoff.

## 2. Material and methods

The full methodology of TITAN, including detailed inclusion criteria, has been published [22,23]. Briefly, patients with mCSPC, ≥1 distant metastases on conventional imaging and Eastern Cooperative Oncology Group performance status 0 or 1 were included. TITAN was conducted at 260 sites in 23 countries in accordance with current International Council for Harmonisation guidelines for Good Clinical Practice and according to the principles of the Declaration of Helsinki. Institutional review boards approved the study, and patients provided written informed consent. Dual primary end-points were rPFS and OS; exploratory end-points included time to PSA progression and PFS2. Additional details can be found in the Supplementary Methods.

These protocol-defined and *post hoc* analyses assessed efficacy in the intent-to-treat (ITT) populations using rPFS (time to radiographic progressive disease or death from any cause), OS (time from randomisation to the date of death from any cause), PFS2 (time from randomisation to first occurrence of disease progression on first subsequent therapy for prostate cancer or death), confirmed PSA decline (decline of ≥50% or ≥90% from baseline or decline to ≤0.2 ng/mL achieved at any time during the study and confirmed ≥4 weeks later), time to PSA progression (time from date of randomisation to date of PSA progression based on Prostate Cancer Working Group 2 (PCWG2) criteria [27]) and time to castration resistance (time from date of randomisation to date of PSA progression, based on PCWG2 criteria [27]). Imaging was assessed per Response Evaluation Criteria In Solid Tumors, version 1.1, modified by PCWG2 criteria [27]. Deep PSA decline was defined as either ≥90% PSA decline or PSA ≤0.2 ng/mL.

Treatment effect of apalutamide was analysed by disease volume, synchronous/metachronous metastases or both. High-volume disease was defined as either visceral metastases with ≥1 bone lesion or ≥4 bone lesions including one outside of the vertebral column or pelvis (modified CHAARTED criteria [16]). Patients who did not meet the high-volume disease criteria were considered to have low-volume disease. Synchronous disease was defined as metastases at initial diagnosis; metachronous disease was defined as metastases developed after localised disease. The effect of the number of metastatic lesions was assessed in apalutamide-treated patients with oligometastases (≤5 in bone only) versus those with polymetastases (> 5 in bone with/without other locations or ≤5 in bone and other locations). The subgroup analyses by volume (high or low) and timing of metastasis presentation (synchronous or metachronous) were defined in the TITAN protocol; the other subgroup analyses were *post hoc*.

Treatment-emergent adverse events (TEAEs), defined as AEs occurring on or after the first dose of the study drug through 30 days after the last study treatment, were graded according to National Cancer Institute Common Terminology Criteria for Adverse Events version 4.0.3 to grade 1 or mild (symptoms that were easily tolerated, causing minimal discomfort and not interfering with everyday activities), grade 2 or moderate (sufficient discomfort was present to cause interference with normal activity), grade 3 or severe (extreme distress, causing significant impairment of functioning or incapacitation; prevents normal everyday activities), grade 4 or life-threatening (urgent intervention indicated) and grade 5 or death.

A TEAE is considered a serious adverse event (SAE) based on International Council for Harmonisation and European Union Guidelines on Pharmacovigilance for Medicinal Products for Human Use if, at any dose, it results in death or in persistent or significant disability/incapacity, requires inpatient hospitalisation or prolongation of existing hospitalisation, or is life-threatening, a congenital anomaly/birth defect, a suspected transmission of any infectious agent via a medicinal product or medically important. A medically important event may be considered an SAE if it jeopardises the patient or may require intervention to prevent the above outcomes despite not necessarily being immediately life-threatening or resulting in death or hospitalisation.

### 2.1. Statistical analysis

Demographic and clinical characteristics were summarised descriptively. rPFS was assessed using first interim analysis cutoff that was prespecified to be final. OS and other clinical end-points were assessed at the final analysis cutoff that analysed crossover patients as a part of the ITT population in the placebo group. The sample size determination is described in the Supplementary Methods. All time-to-event

UNCORRECTED PROOF

estimates for subgroups were estimated using the Kaplan-Meier method and the Cox proportional-hazards model to assess hazard ratios and associated 95% confidence intervals (CIs). Nominal $p$ values were reported without adjustment for multiplicity. Two-sided $p$ values of $< 0.05$ were considered statistically significant. TEAEs were assessed in the safety populations (patients who received the study drug) using the final analysis cutoff and were summarised descriptively or using cumulative incidences.

## 3. Results

### 3.1. Demographics and disease characteristics at baseline

A total of 1052 patients were randomised in a 1:1 ratio to receive either 240 mg/day apalutamide ($n = 525$) or placebo ($n = 527$), plus ongoing ADT, between 15th December 2015 and 25th July 2017 (Fig. S1). The prespecified first interim analysis (final analysis for rPFS) occurred at the cutoff on 23rd November 2018, after the number of required events occurred within 22.7 mo of follow-up. The prespecified final analysis occurred at the cutoff on 7th September 2020 after the number of required deaths had occurred within 44.0 mo of follow-up.

The majority of TITAN patients had high-volume (63%; 660/1052) and synchronous (81%; 852/1052) disease. Four subgroups stratified by disease volume and timing of metastasis presentation comprised 54% (567/1052; synchronous/high-volume), 27% (285/1052; synchronous/low-volume disease), 5.7% (60/1052; metachronous/high-volume), and 8.0% (84/1052; metachronous/low-volume) of patients. Major disease characteristics pertinent to this analysis were balanced across treatment groups and subgroups (Table 1). Bone-only metastases were present in 558 (53%) patients, including 286 (27%) with ≤5 and 272 (26%) with > 5 lesions. Metastases in bone plus visceral locations were present in 128 (12%) patients. Only 11% (58) of apalutamide-treated patients had up to six cycles of docetaxel prior to apalutamide plus ADT; of those, 44 patients had high-volume disease and 40 had high-volume/synchronous disease. Of patients with high-volume disease, 81% (532/660) had > 5 bone lesions, and of patients with low-volume disease, 91% (356/392) had ≤5 bone lesions.

### 3.2. Apalutamide treatment effect across subgroups

In general, apalutamide plus ADT improved outcomes in all patient subgroups compared with ADT alone. In the synchronous/high-volume subgroup, apalutamide improved OS with nominal $p$-values $< 0.05$ (Fig. 1A; HR = 0.68 [95% CI: 0.53–0.87]). The OS event rates (number of events over patients at risk) in this subgroup for apalutamide- and placebo-treated patients were 112/273 (41%) and 156/294 (53%) (Table S1). In the synchronous/low-volume subgroup, the benefit in OS also favoured apalutamide but nominal $p$-value was $> 0.05$ (Fig. 1B; HR = 0.65 [95% CI: 0.40–1.05]); the event rates were 28/138 (20%) and 43/147 (29%) for apalutamide- and placebo-treated patients (Table S1). The treatment effect in OS

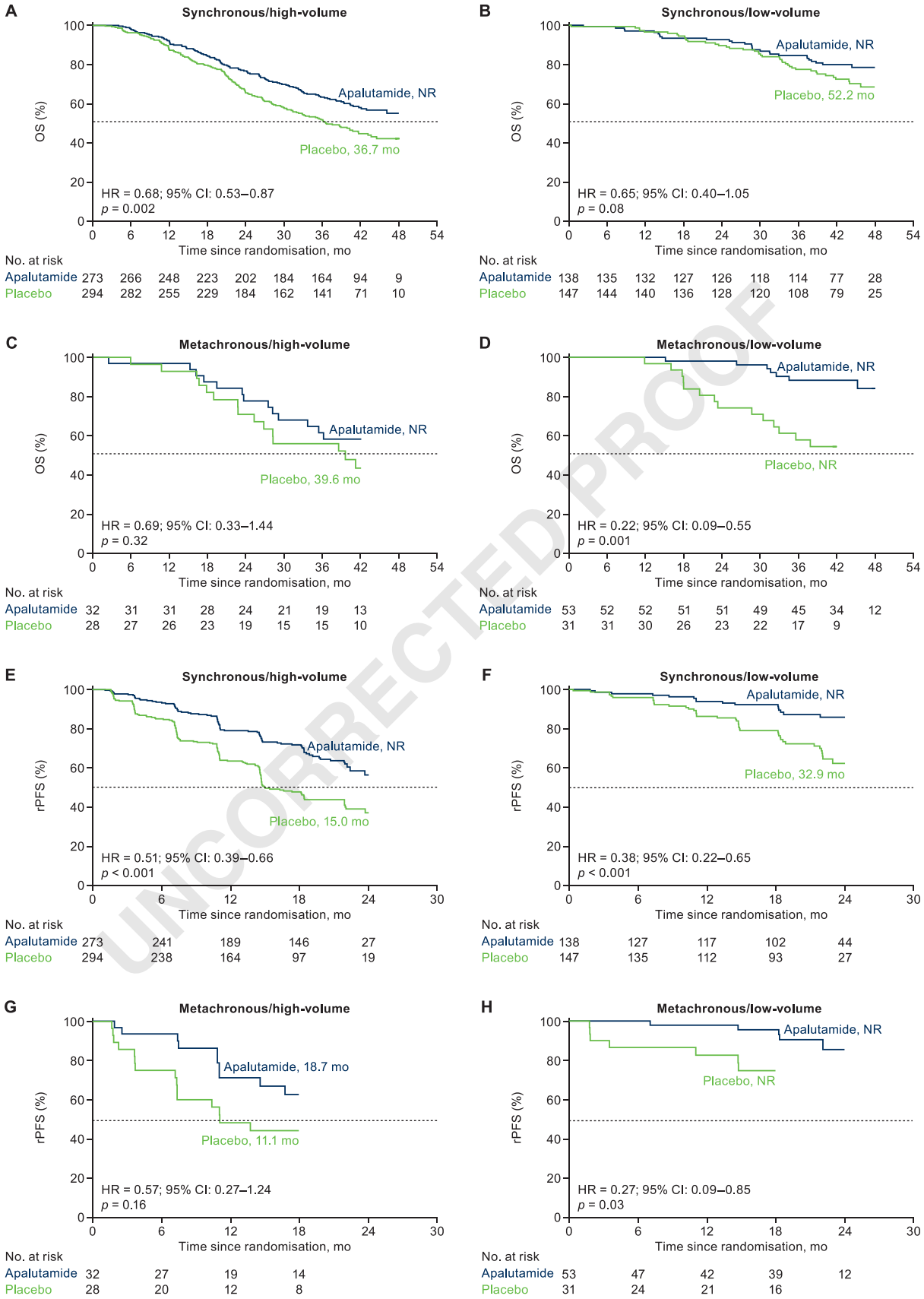
Table 1
Baseline demographics and disease characteristics (intent-to-treat (ITT) population). All data are reported as n (%) unless otherwise stated.

| | Apalutamide + ADT ($n = 525$) | Placebo + ADT ($n = 527$) |
|---|---|---|
| Median age in years (range) | 69 (45–94) | 68 (43–90) |
| Race | | |
| White | 354 (67%) | 365 (69%) |
| Asian | 119 (23%) | 110 (21%) |
| Black or African American | 10 (1.9%) | 9 (1.7%) |
| American Indian or Alaska native | 6 (1.1%) | 13 (2.5%) |
| Not reported | 11 (2.1%) | 8 (1.5%) |
| Other | 24 (4.6%) | 22 (4.2%) |
| Multiple | 1 (0.2%) | 0 |
| Gleason score at initial diagnosis | | |
| ≤7 | 174 (33%) | 169 (32%) |
| > 7 | 351 (67%) | 358 (68%) |
| ECOG PS score | | |
| 0 | 328 (63%) | 348 (66%) |
| 1 | 197 (38%) | 178 (34%) |
| 2 | 0 | 1 (0.2%) |
| Disease volume | | |
| High | 325 (62%) | 335 (64%) |
| Low | 200 (38%) | 192 (36%) |
| Previous docetaxel treatment | 58 (11%) | 55 (10%) |
| Median PSA, ng/mL (range) | 5.97 (0–2682) | 4.02 (0–2229) |
| Metastatic presentation at initial diagnosis | | |
| Synchronous | 411 (78%) | 441 (84%) |
| Synchronous/high-volume | 273 (52%) | 294 (56%) |
| Synchronous/low-volume | 138 (26%) | 147 (28%) |
| Metachronous | 85 (16%) | 59 (11%) |
| Metachronous/high-volume | 32 (6.1%) | 28 (5.3%) |
| Metachronous/low-volume | 53 (10%) | 31 (5.9%) |
| MX | 29 (5.5%) | 27 (5.1%) |
| Most common (≥5%) mutually exclusive anatomical locations of metastases at study entry | | |
| Bone-only | 289 (55%) | 269 (51%) |
| Bone-only ≤5 lesions | 150 (29%) | 136 (26%) |
| Bone-only > 5 lesions | 139 (27%) | 133 (25%) |
| Bone ≤5 lesions + other locations | 89 (17%) | 109 (21%) |
| Bone > 5 lesions + other locations | 147 (28%) | 149 (28%) |
| Bone + lymph node | 163 (31%) | 166 (32%) |
| Bone + viscera | 24 (4.6%) | 30 (5.7%) |
| Bone + lymph node + viscera | 27 (5.1%) | 35 (6.6%) |
| Bone + visceral + other locations (soft tissue or lymph node) | 5 (1.0%) | 7 (1.3%) |
| Bone + other locations[a] | 180 (34%) | 186 (35%) |
| Mean baseline BPI-SF pain score | | |
| 0: no pain | 198 (38%) | 200 (38%) |
| 1–3: mild pain | 195 (37%) | 207 (39%) |
| 4–7: moderate pain | 98 (19%) | 95 (18%) |
| 8–10: severe pain | 12 (2.3%) | 11 (2.1%) |
| Missing data | 22 (4.2%) | 14 (2.7%) |

ADT: androgen-deprivation therapy; BPI-SF: Brief Pain Inventory (Short Form); ECOG PS: Eastern Cooperative Oncology Group performance status; MX: metastasis cannot be measured; PSA: prostate-specific antigen.

[a] Includes bone + lymph node, bone + soft tissue and bone + lymph node + soft tissue.

UNCORRECTED PROOF




UNCORRECTED PROOF

Fig. 1. Treatment effect of apalutamide added to androgen-deprivation therapy (ADT) on outcomes in patients with different disease volume or timing of metastasis occurrence. Kaplan-Meier plots for overall survival (OS) (A–D) and radiographic progression-free survival (rPFS) (E–H) and forest plots for second progression-free survival (PFS2) and time to castration resistance (I) were analysed in patients with synchronous/high-volume, synchronous/low-volume, metachronous/high-volume, and metachronous/low-volume disease. The Kaplan-Meier curves are truncated for instances in which the number at risk in a group was < 5. NR: not reached.

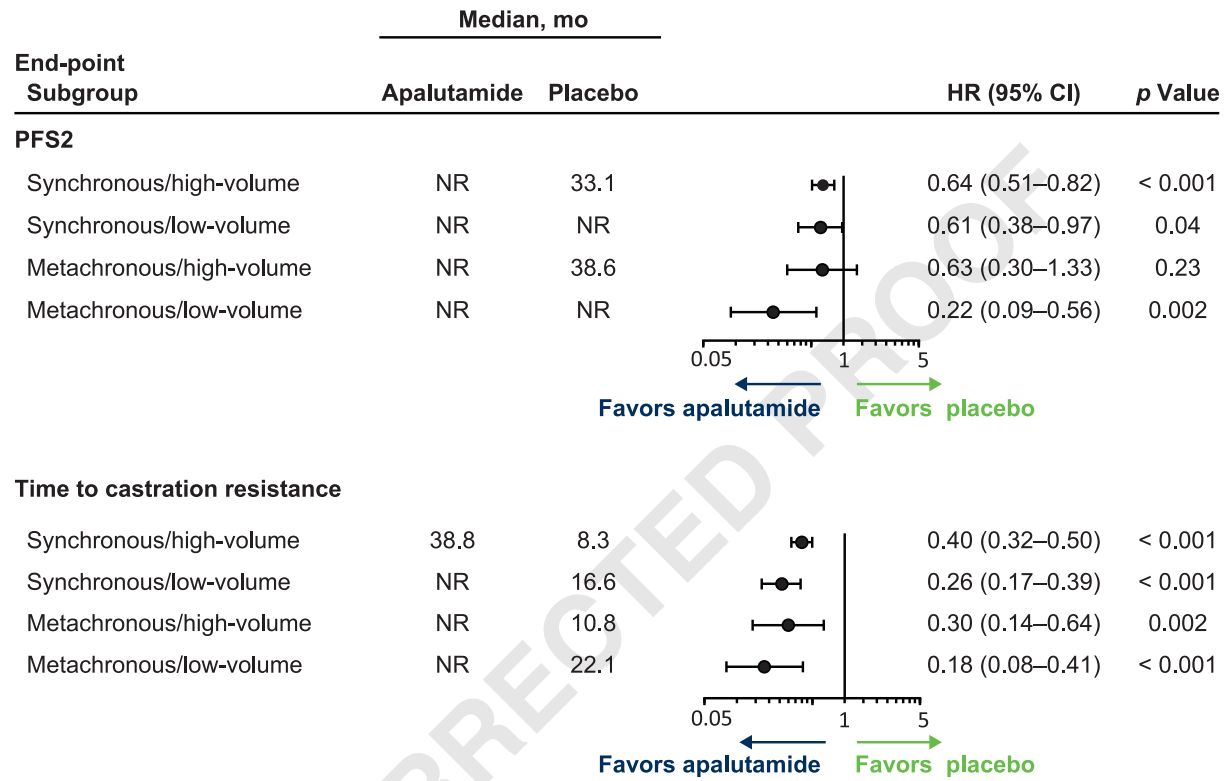



Fig. 1. (*continued*)

UNCORRECTED PROOF

favoured apalutamide with nominal *p*-value > 0.05 in the metachronous/high-volume subgroup (Fig. 1C; HR = 0.69 [95% CI: 0.33–1.44]) and with *p*-value < 0.05 in the metachronous/low-volume subgroup (Fig. 1D; HR = 0.22 [95% CI: 0.09–0.55]). Corresponding event rates for apalutamide- and placebo-treated patients were 13/32 (41%) and 15/28 (54%) in the metachronous/high-volume and 7/53 (13%) and 14/31 (45%) in metachronous/low-volume subgroups.

Apalutamide treatment effect was observed despite more placebo-treated patients receiving first life-prolonging subsequent therapy for prostate cancer (72% [212/294] in the synchronous/high-volume, 54% [80/147] in synchronous/low-volume, 75% [21/28] in metachronous/high/volume, and 65% [20/31] in metachronous/low-volume subgroups) than apalutamide-treated patients (56% [153/273] in the synchronous/high-volume, 28% [39/138] in synchronous/low-volume, 56% [18/32] in metachronous/high/volume and 43% [23/53] metachronous/low-volume subgroups) (Table 2).

Benefit in rPFS was generally consistent with that in OS. Thus, in the synchronous/high-volume and synchronous/low-volume subgroups, apalutamide improved rPFS with nominal *p*-value < 0.05 (Fig. 1E-F; HR = 0.51 [95% CI: 0.39–0.66] and 0.38 [0.22–0.65]). Benefit in rPFS favoured apalutamide with nominal *p*-value > 0.05 in the metachronous/high-volume subgroup (Fig. 1G; HR = 0.57 [95% CI: 0.27–1.24]) and with *p*-value < 0.05 in the metachronous/low-volume subgroup (Fig. 1H; HR = 0.27 [95% CI: 0.09–0.85]). rPFS event rates for apalutamide- and placebo-treated patients were 89/273 (33%) and 151/294 (51%), 19/138 (14%) and 45/147 (31%), 12/32 (38%) and 15/28 (54%), and 5/53 (9.4%) and 8/31 (26%), in synchronous/high-volume, synchronous/low-volume, metachronous/high-volume, and metachronous/low-volume subgroups, respectively (Table S1).

Consistent with findings for the dual primary end-points, addition of apalutamide improved PFS2 (Fig. 1I, Fig. S2) in all subgroups with nominal *p*-value < 0.05, except metachronous/high-volume, and prolonged time to castration resistance (Fig. 1I, Fig. S3) and to PSA progression (Fig. S4A-D) in all four subgroups. The addition of apalutamide also increased the proportion of patients with confirmed best ≥50% PSA decline or deep PSA decline of ≥90% or to ≤0.2 ng/mL at any time during the study, regardless of stratification (Fig. S5).

Table 2
Select first subsequent life-prolonging therapy for prostate cancer after study treatment discontinuation. All data are reported as n (%).

| | Synchronous/High-volume | | Synchronous/Low-volume | | Metachronous/High-volume | | Metachronous/Low-volume | |
|---|---|---|---|---|---|---|---|---|
| | Apalutamide + ADT ($n = 273$) | Placebo + ADT ($n = 294$) | Apalutamide + ADT ($n = 138$) | Placebo + ADT ($n = 147$) | Apalutamide + ADT ($n = 32$) | Placebo + ADT ($n = 28$) | Apalutamide + ADT ($n = 53$) | Placebo + ADT ($n = 31$) |
| Patients who discontinued treatment and remained alive (denominator for below), n | 153 | 212 | 39 | 80 | 18 | 21 | 23 | 20 |
| Abiraterone acetate plus prednisone | 26 (17%) | 60 (28%) | 7 (18%) | 22 (28%) | 4 (22%) | 6 (29%) | 2 (8.7%) | 5 (25%) |
| Enzalutamide | 11 (7.2%) | 28 (13%) | 3 (7.7%) | 18 (23%) | 2 (11%) | 5 (24%) | 2 (8.7%) | 4 (20%) |
| Docetaxel | 43 (28%) | 70 (33%) | 5 (13%) | 26 (33%) | 3 (17%) | 6 (29%) | 3 (13%) | 4 (20%) |
| Cabazitaxel | 11 (7.2%) | 22 (10%) | 2 (5.1%) | 7 (8.8%) | 1 (5.6%) | 0 | 1 (4.3%) | 0 |
| Radium-223 | 8 (5.2%) | 13 (6.1%) | 2 (5.1%) | 0 | 1 (5.6%) | 2 (9.5%) | 1 (4.3%) | 1 (5.0%) |
| Sipuleucel-T | 1 (0.7%) | 3 (1.4%) | 2 (5.1%) | 1 (1.3%) | 0 | 2 (9.5%) | 1 (4.3%) | 1 (5.0%) |

ADT: androgen-deprivation therapy.

When the outcomes were assessed in subgroups stratified by one factor to increase the number of events (Table S1), improvement of all outcomes with apalutamide plus ADT had nominal $p$-values $< 0.05$ in both synchronous/metachronous (Fig. S4E-F, Fig. S6) and high-/low-volume subgroups (Fig. S4G-H, Fig. S7) compared with ADT alone. Apalutamide-treated patients from the metachronous subgroup or from the low-volume subgroup had improved OS compared with those from synchronous or high-volume subgroups, respectively: HR =0.62 [95%CI: 0.39–0.997], $p = 0.046$ and 0.36 [0.25–0.51], $p < 0.001$. This benefit was seen for the other outcomes (data not shown). The treatment effect (apalutamide plus ADT versus ADT alone) for OS and rPFS was homogeneous across high- and low-volume (interaction effect $p$-values 0.262 and 0.194 for OS and rPFS, respectively) or across metachronous synchronous subgroups (interaction effect $p$-values 0.075 and 0.526, respectively).

### 3.3. Effect of the number and location of metastases on outcomes in apalutamide-treated patients

Apalutamide-treated patients with oligometastatic disease with ≤5 bone-only metastases had the most favourable OS and other outcomes versus those with polymetastatic disease with either > 5 bone-only metastases, > 5 bone metastases plus metastases in other organs or ≤5 bone metastases plus additional metastases in other organs (Fig. 2). Poorest outcomes were observed in patients with polymetastatic disease with > 5 bone metastases plus metastases in other locations. Compared with patients with bone-only disease ($n = 289$), those with metastases in bone plus visceral plus other locations ($n = 56$) or those with bone plus metastases in other locations ($n = 180$) had shorter OS (HR = 2.63 [95% CI: 1.68–4.10] and 1.90 [1.37–2.63]) and rPFS (3.42 [2.10–5.55] and 2.22 [1.52–3.25]; $p < 0.001$ for all). Similar results were observed for other outcomes (data not shown).

In placebo-treated patients, the outcomes generally followed the pattern in apalutamide-treated patients, except for patients with ≤5 bone metastases plus other locations, in whom results were inconclusive (Fig. S8).

### 3.4. Safety

In the high-volume subgroup, TEAEs, grade 3–4 TEAEs and SAEs occurred in 96%, 54% and 29% of apalutamide-treated patients and in 97%, 47% and 22% of placebo-treated patients, respectively. In the low-volume subgroup, TEAEs, grade 3–4 TEAEs and SAEs occurred in 99%, 43%, and 30% of apalutamide- and in 96%, 33% and 21% of placebo-treated patients, respectively. Similar rates of TEAEs, grade 3–4 TEAEs, SAEs and TEAEs of interest were observed across treatment groups in metachronous and synchronous subgroups, except that falls and fractures occurred in more patients from the metachronous than synchronous subgroups. Cumulative incidences of grade

UNCORRECTED PROOF

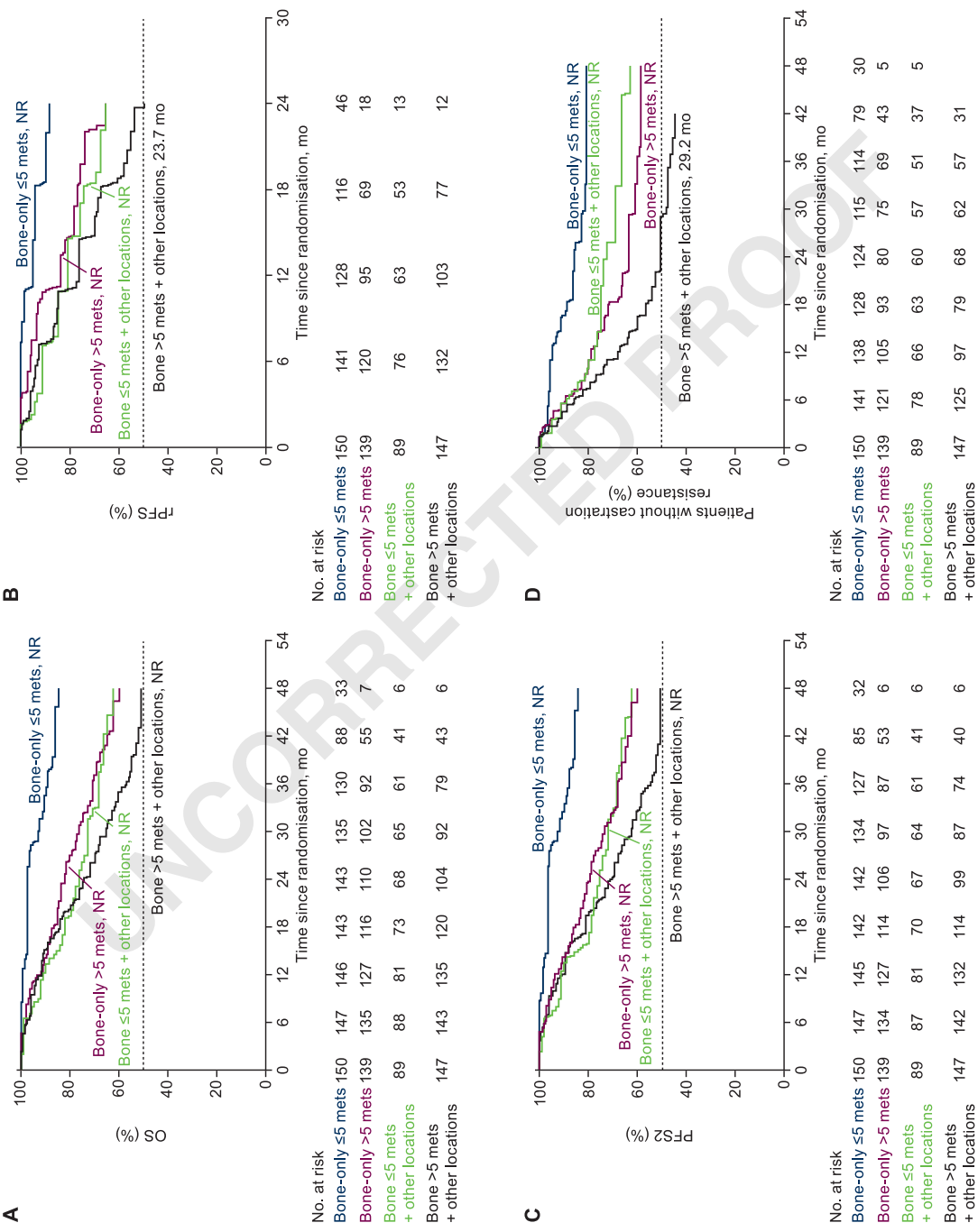



Fig. 2. The effect of oligo- and polymetastatic disease on clinical outcomes following apalutamide plus ADT treatment. OS (A), rPFS (B), PFS2 (C) and time to castration resistance (D). The curves are truncated for instances in which the number at risk in a group was < 5 metastases. Mets: metastases; NR: not reached.

UNCORRECTED PROOF

3–4 (Fig. 3) and serious TEAEs (data not shown) were similar in treatment groups regardless of disease volume or metastatic presentation at diagnosis.

## 4. Discussion

Apalutamide added to concurrent ADT resulted in clinical benefits in the overall TITAN population [22,23]. These protocol-defined and *post hoc* analyses of TITAN patients with mCSPC confirm apalutamide benefit across a range of clinical prognostic subgroups with varying disease characteristics based on disease volume, timing of metastatic presentation at diagnosis and number of metastases. Deep PSA decline and consistent efficacy across all outcomes were also observed with apalutamide irrespective of disease volume or timing of metastasis occurrence. Although patients with low-volume or metachronous disease had the greatest benefit with apalutamide plus ADT, patients from other subgroups, including those with synchronous/high-volume disease, a subgroup that has poor survival prognosis with ADT alone [1], also derived benefit from apalutamide plus ADT. Our findings are consistent with deep PSA decline observed in the majority of the TITAN population which translated into improved long-term outcomes [28–30]. The safety profile of apalutamide across subgroups remained consistent with previous reports [22,23]. These data support the efficacy and safety of treatment intensification with apalutamide in the diverse population of patients with mCSPC.

We observed clinical benefit in terms of estimated HR favouring apalutamide across all subgroups, although nominal *p*-values for OS, rPFS and PFS2 in the metachronous/high-volume subgroup or for OS in the synchronous/low-volume subgroup were $>0.05$. The small number of patients and events in these subgroups decreased the statistical power, and therefore the conclusions for these subgroups should be confirmed in studies with larger numbers of patients and longer follow-ups. However, HR values for OS in metachronous/high-volume and synchronous/low-volume subgroups were very similar to those in the synchronous/high-volume subgroup, supporting the consistency of benefit across three subgroups. The lack of interaction with high-/low-volume or synchronous/metachronous disease presentation further supports the consistent treatment effect of apalutamide.

Similar subgroup analyses from ARCHES assessed the efficacy of enzalutamide and ADT in mCSPC [10,11,31]. Those analyses also showed that, compared with ADT alone, enzalutamide added to ADT improved rPFS and other secondary end-points regardless of volume, oligometastatic/polymetastatic disease, metachronous/synchronous metastases or pattern of metastatic spread.

Treatment regimens suggested for patients with mCSPC depend on disease burden, and associated risks are still being investigated. Among the four subgroups stratified by timing of metastasis presentation and disease volume, patients with metachronous/low-volume mCSPC had the lowest and those with synchronous/high-volume disease had the highest risk of death or developing castration resistance [1]. The metachronous/low-volume subgroup did not benefit from doublet treatment with docetaxel and ADT compared with ADT alone [19] but did from the enzalutamide with ADT combination compared with nonsteroidal antiandrogen and ADT. Of note, the latter benefit was observed in ENZAMET patients [3,21], approximately half of whom received concurrent docetaxel as part of their standard of care [3,21,32]. We observed a treatment effect of the apalutamide and ADT combination compared with ADT alone with nominal two-sided $p < 0.05$ across all low-volume subgroups, including those with metachronous or synchronous characteristics.

Doublet or triplet combination treatments with docetaxel and ADT or an AR pathway synthesis inhibitor appear to be most effective in poor-prognosis high-volume subgroups. The meta-analysis of docetaxel and ADT studies showed that the docetaxel effect is greater with higher volume and bulkiness of disease, as defined by increasing T-stage, and, to a lesser degree, with synchronous metastases [19]. Triplet treatment with docetaxel, ADT and AAP or docetaxel, ADT and darolutamide is now recommended for patients with mCSPC who are fit for docetaxel based on the PEACE-1 and ARASENS studies, respectively [4,33,34]. However, patient populations that would benefit from triplet therapy still need to be defined. The updated subgroup analysis of ARASENS did not show significantly improved OS in metachronous/high-volume disease with triplet treatment [20]. We observed a treatment effect of apalutamide based on hazard ratio and *p*-value in the synchronous/high-volume subgroup across all assessed end-points. The efficacy of triplet apalutamide, docetaxel and ADT treatment is the subject of future studies; however, we have shown that OS and other clinical outcomes in the overall and case-matched TITAN populations and in the high-volume subpopulation were similar regardless of prior docetaxel [35]. As the synchronous/high-volume subgroup represents 91% (40/44) of the high-volume population, similar results by prior docetaxel in this subgroup would likely be observed. Whether concurrent docetaxel provides additional benefit to apalutamide plus ADT must be assessed prospectively.

The limitations of this analysis include the *post hoc* nature of some of these analyses, the relatively small size of some subgroups and the relatively short follow-up of the better-prognosis synchronous/low-volume subgroup, which limited interpretability. Metastatic burden was defined by conventional imaging only; use of more sensitive measures may have affected the results. Additional prospective analyses of large patient populations are needed to confirm our findings.

UNCORRECTED PROOF

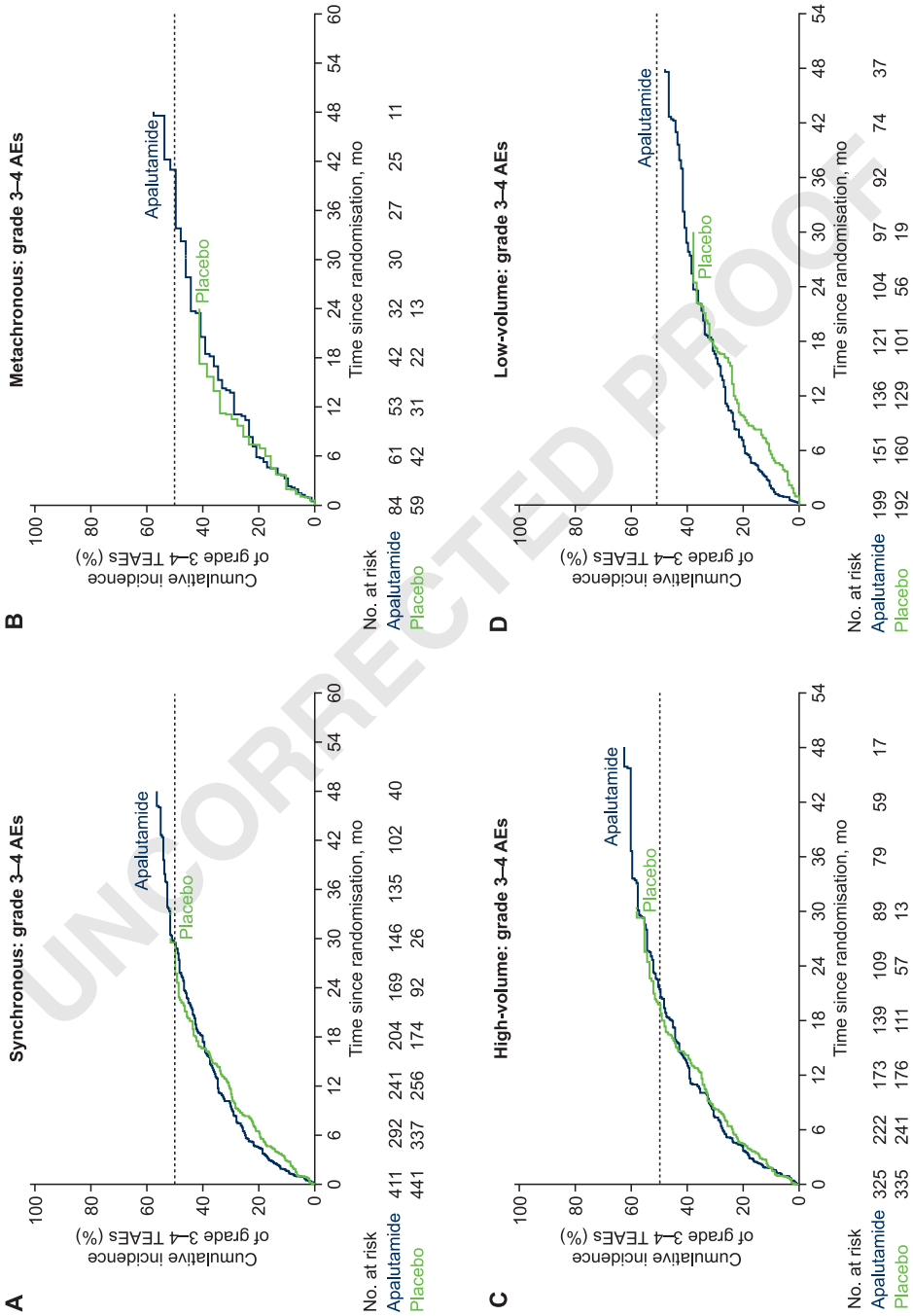



Fig. 3. Cumulative incidences of grade 3–4 treatment-emergent adverse events (TEAEs) in apalutamide plus ADT and placebo plus ADT treatment groups of TITAN. Synchronous (A), metachronous (B), high-volume (C) and low-volume (D) disease. AE: adverse event.

UNCORRECTED PROOF

## 5. Conclusions

This protocol-defined and *post hoc* analysis of TITAN patients with mCSPC showed that apalutamide plus ADT produced a robust long-term benefit regardless of stratification by disease volume, synchronous/metachronous metastases, or combination of both factors. Among patients treated with apalutamide, those with low-volume disease or metachronous metastases derived greater benefits than those with high-volume disease or synchronous metastases according to the hazard ratio. However, in patients with synchronous/high-volume disease, the addition of apalutamide to ADT also resulted in significantly improved survival and other clinical outcomes. Our findings are consistent with a robust benefit/risk profile of apalutamide in mCSPC, supporting effective and safe early treatment in all subgroups of patients with mCSPC.

## Role of the funding source

The sponsor (Janssen Research & Development) was involved in the design and conduct of the study; collection, management, analysis and interpretation of the data; and preparation and review of the manuscript. All the authors had access to the data, drafted the manuscript with input from the sponsor (Janssen), reviewed and approved the manuscript before submission and made the decision to submit the manuscript for publication.

## Funding

This work was supported by Janssen Research & Development.

## CRediT authorship contribution statement

**Axel S. Merseburger:** Conceptualization, Methodology, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Neeraj Agarwal:** Conceptualization, Methodology, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Laurence I. Karsh:** Conceptualization, Methodology, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Andrea J. Pereira de Santana Gomes:** Conceptualization, Methodology, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Álvaro Juárez Soto:** Conceptualization, Methodology, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Robert W. Given:** Conceptualization, Methodology, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Hirotsugu Uemura:** Conceptualization, Methodology, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Simon Chowdhury:** Conceptualization, Methodology, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Kim N. Chi:** Conceptualization, Methodology, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Anders Bjartell:** Conceptualization, Methodology, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Amitabha Bhaumik:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing. **Florence Lefresne:** Conceptualization, Methodology, Investigation, Resources, Supervision, Writing – original draft, Writing – review & editing. **Sabine D. Brookman-May**: Conceptualization, Methodology, Investigation, Resources, Supervision, Writing – original draft, Writing – review & editing. **Suneel D. Mundle:** Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing. **Sharon A. McCarthy:** Conceptualization, Methodology, Investigation, Resources, Supervision, Writing – original draft, Writing – review & editing.

## Declaration of Competing Interest

Dr. Axel Merseburger reports consulting and advisory role for Astellas, AstraZeneca, Bristol-Myers Squibb, EUSAPharm, Ipsen, Janssen, Merck Serono, MSD, Novartis, Teva, Pfizer, Roche and Takeda; lectures/speaker/honoraria for/from Astellas, AstraZeneca, Bristol-Myers Squibb, Eisai, Ipsen, Janssen, Merck Serono, MSD, Novartis, Pfizer, Roche, Takeda and Teva; research and clinical studies for Astellas, AstraZeneca, Bristol-Myers Squibb, EUSAPharm, Ipsen, Janssen, Merck Serono, MSD, Novartis, Pfizer, Roche, Takeda and Teva.

Dr. Neeraj Agarwal reports consulting and advisory role for Astellas, AstraZeneca, Aveo, Bayer, Bristol Myers Squibb, Calithera, Clovis, Eisai, Eli Lilly, EMD Serono, Exelixis, Foundation Medicine, Genentech, Gilead, Janssen, Merck, MEI Pharma, Nektar, Novartis, Pfizer, Pharmacyclics and Seattle Genetics; and research funding to institution from Astellas, AstraZeneca, Bavarian Nordic, Bayer, Bristol Myers Squibb, Calithera, Celldex, Clovis, Eisai, Eli Lilly, EMD Serono, Exelixis, Genentech, Gilead, Glaxo Smith Kline, Immunomedics, Janssen, Medivation, Merck, Nektar, New Link Genetics, Novartis, Pfizer, Prometheus, Rexahn, Roche, Sanofi, Seattle Genetics, Takeda and Tracon.

Dr. Laurence Karsh reports consulting or advisory role for 3D Biopsy, Astellas, AstraZeneca, Bayer, Dendreon, Ferring, Janssen, Pfizer and Vaxiion; speakers' bureau for Astellas, Bayer, Janssen, Pfizer and Clovis; honoraria from Astellas, Bayer, Janssen, Pfizer and Dendreon; stock or other ownership from Swan Valley Medical; research funding from Astellas, AstraZeneca, Bayer, BioXcel Therapeutics, Bristol Meyers Squibb, BU Optics, CUSP, Dendreon, Epizyme, Exact Sciences, Ferring, FKD, Genetech/Roche, GenomeDX, Genomic Health, Janssen, Merck, Myovant, Nucleix, OncoCell MDx, Pfizer, Pharmtech/

UNCORRECTED PROOF

Very, Precision Med and QED; and travel, accommodations and expenses from Astellas, Bayer, Dendreon, Janssen and Pfizer.

Dr. Andrea Juliana Pereira de Santana Gomes reports consulting and advisory role for Janssen Oncology; speakers' bureau for Astellas Pharma, Bayer and Janssen Oncology; research funding from Bayer, Janssen Oncology, MSD Oncology and Roche.

Dr. Alvaro Juarez Soto reports consulting and advisory role for Astellas, Bayer and Janssen; meeting participation/lecturer for Astellas, Bayer and Janssen; investigator for Bayer and Janssen; meeting participation/lecturer for Astellas, Bayer and Janssen; clinical study participation for Janssen.

Dr. Robert Given reports speakers' bureau for Bayer, Janssen and Myovant; clinical study participation for Bayer and Janssen.

Dr. Hirotsugu Uemura reports consulting and advisory role for AstraZeneca, Bayer, Janssen, MSD, Ono/BMS and Sanofi; meeting participation/lecturer for Astellas, AstraZeneca, Bayer, Janssen, MSD, Novartis, Ono/BMS, Pfizer and Takeda; investigator for Astellas, AstraZeneca, Bayer, Janssen, MSD, Ono/BMS, Pfizer, Sanofi, Taiho and Takeda; clinical study participation for Bayer, Janssen, MSD and Ono/BMS.

Dr. Simon Chowdhury reports consulting and advisory role to Astellas Pharma, Bayer, Beigene, Clovis, Janssen-Cilag, Johnson & Johnson, Novartis and Sanofi; research funding (self) from Clovis; and honoraria from Astellas Pharma, Bayer, Beigene, Clovis, Janssen-Cilag, Johnson & Johnson, Novartis and Sanofi.

Dr. Kim Chi reports consulting and advisory role to Janssen, Astellas, Essa, Sanofi, Bayer, Roche, AstraZeneca, Daiichi Sankyo, Novartis, Pfizer, Merck, Bristol Myers Squibb and Point Pharma; research funding to institution from Janssen, Astellas and Sanofi; and honoraria from Janssen, Astellas, Bayer, AstraZeneca, Merck and Pfizer.

Dr. Anders Bjartell reports consulting and advisory role for Astellas and Bayer; board membership/officer/trustee for Glactone Pharma and LIDDS Pharma; meeting participation/lecturer for Astellas, AstraZeneca, Bayer, Incyte, Ipsen, Janssen and Merk; clinical study participation for Astellas, Ferring and Janssen; stock ownership in Glactone Pharma, LIDDS Pharma and WntResearch.

Dr. Amitabha Bhaumik, Dr. Florence Lefresne, Dr. Sabine Brookman-May, Dr. Suneel Mundle and Ms. Sharon McCarthy are employees of Janssen Research & Development and may hold stock in Johnson & Johnson. Dr. Amitabha Bhaumik also reports stock ownership of AbbVie.

## Acknowledgements

The authors would like to thank the patients who participated in this study and their families, as well as the investigators, study coordinators, study teams, and nurses.

Writing assistance was provided by Larissa Belova, PhD, of Parexel, and was funded by Janssen Global Services, LLC.

## Submission declaration

The results of these analyses have been partially presented at EAU20, EAU21 and AUA21.

## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ejca.2023.113290.

## References

[1] Francini E, Gray KP, Xie W, Shaw GK, Valenca L, Bernard B, et al. Time of metastatic disease presentation and volume of disease are prognostic for metastatic hormone sensitive prostate cancer (mHSPC). Prostate 2018;78(12):889–95. https://doi.org/10.1002/pros.23645.

[2] Gillessen S, Attard G, Beer TM, Beltran H, Bjartell A, Bossi A, et al. Management of patients with advanced prostate cancer: report of the advanced prostate cancer consensus conference 2019. Eur Urol 2020;77(4):508–47. https://doi.org/10.1016/j.eururo.2020.01.012.

[3] Sweeney CJ, Martin AJ, Stockler MR, Begbie S, Chi KN, Chowdhury S, et al. Overall survival of men with metachronous metastatic hormone-sensitive prostate cancer treated with enzalutamide and androgen deprivation therapy. Eur Urol 2021;80(3):275–9. https://doi.org/10.1016/j.eururo.2021.05.016.

[4] Fizazi K, Foulon S, Carles J, Roubaud G, McDermott R, Flechon A, et al. Abiraterone plus prednisone added to androgen deprivation therapy and docetaxel in de novo metastatic castration-sensitive prostate cancer (PEACE-1): a multicentre, open-label, randomised, phase 3 study with a 2×2 factorial design. Lancet 2022;399(10336):1695–707. https://doi.org/10.1016/S0140-6736(22)00367-1.

[5] Gillessen S, Bossi A, Davis ID, de Bono J, Fizazi K, James ND, et al. Management of patients with advanced prostate cancer. Metastatic and/or castration resistant prostate cancer: report of the Advanced Prostate Cancer Consensus Conference (APCCC) 2022. Eur J Cancer 2023. https://doi.org/10.1016/j.ejca.2023.02.018.

[6] Bossi A, Foulon S, Maldonado X, Sargos P, McDermott RS, Flechon A, et al. Prostate irradiation in men with de novo, low-volume, metastatic, castration-sensitive prostate cancer (mCSPC): results of PEACE-1, a phase 3 randomized trial with a 2×2 design. Presented at: 2023 ASCO Annual Meeting, Chicago, IL, USA; 2023.

[7] Armstrong AJ, Szmulewitz RZ, Petrylak DP, Holzbeierlein J, Villers A, Azad A, et al. ARCHES: a randomized, phase III study of androgen deprivation therapy with enzalutamide or placebo in men with metastatic hormone-sensitive prostate cancer. J Clin Oncol 2019;37(32):2974–86. https://doi.org/10.1200/JCO.19.00799.

[8] Armstrong AJ, Holzbeierlein J, Iguchi T, Azad A, Villers A, Alekseev B, et al. Society of Urologic Oncology 22nd Annual Meeting. The efficacy of enzalutamide (ENZA) plus androgen deprivation therapy (ADT) on oligometastatic hormone-sensitive prostate cancer: extended post hoc analysis of ARCHES.

UNCORRECTED PROOF

Available from: 〈https://suo-abstracts.secure-platform.com/a/gallery/rounds/12/details/1644〉. [Accessed 24th May 2023].

[9] Armstrong AJ, Iguchi T, Azad A, Szmulewitz RZ, Holzbeierlein J, Villers A, et al. Overall survival (OS) in patients (pts) with metastatic hormone-sensitive prostate cancer (mHSPC) treated with enzalutamide (ENZA) + androgen deprivation therapy (ADT) by high or low disease volume and progression to mHSPC (M0 at diagnosis) or de novo mHSPC (M1 at diagnosis): post hoc analysis of the phase 3 ARCHES trial. J Clin Oncol 2021;40(Suppl.6):115. https://doi.org/10.1200/JCO.2022.40.6_suppl.115.

[10] Armstrong AJ, Iguchi T, Azad A, Villers A, Alekseev B, Petrylak DP, et al. The efficacy of enzalutamide (ENZA) plus androgen deprivation therapy (ADT) on bone oligometastatic hormone-sensitive prostate cancer: a post hoc analysis of ARCHES. J Clin Oncol 2021;39(Suppl.15):5071. https://doi.org/10.1200/JCO.2021.39.15_suppl.5071.

[11] Azad A, Villers A, Alekseev B, Szmulewitz RZ, Alcaraz A, Shore ND, et al. Efficacy of enzalutamide (ENZA) plus androgen deprivation therapy (ADT) in men with de novo (M1) metastatic hormone-sensitive prostate cancer (mHSPC) versus progression to mHSPC (M0): post hoc analysis of the phase III ARCHES trial. J Clin Oncol 2021;39(Suppl.6):102. https://doi.org/10.1200/JCO.2021.39.6_suppl.102.

[12] Armstrong AJ, Azad AA, Iguchi T, Szmulewitz RZ, Petrylak DP, Holzbeierlein J, et al. Improved survival with enzalutamide in patients with metastatic hormone-sensitive prostate cancer. J Clin Oncol 2022;40(15):1616–22. https://doi.org/10.1200/JCO.22.00193.

[13] Azad AA, Armstrong AJ, Alcaraz A, Szmulewitz RZ, Petrylak DP, Holzbeierlein J, et al. Efficacy of enzalutamide in subgroups of men with metastatic hormone-sensitive prostate cancer based on prior therapy, disease volume, and risk. Prostate Cancer Prostatic Dis 2022;25(2):274–82. https://doi.org/10.1038/s41391-021-00436-y.

[14] Gravis G, Boher JM, Chen YH, Liu G, Fizazi K, Carducci MA, et al. Burden of metastatic castrate naive prostate cancer patients, to identify men more likely to benefit from early docetaxel: further analyses of CHAARTED and GETUG-AFU15 studies. Eur Urol 2018;73(6):847–55. https://doi.org/10.1016/j.eururo.2018.02.001.

[15] Gravis G, Boher JM, Joly F, Soulie M, Albiges L, Priou F, et al. Androgen deprivation therapy (ADT) plus docetaxel versus ADT alone in metastatic non castrate prostate cancer: impact of metastatic burden and long-term survival analysis of the randomized phase 3 GETUG-AFU15 trial. Eur Urol 2016;70(2):256–62. https://doi.org/10.1016/j.eururo.2015.11.005.

[16] Kyriakopoulos CE, Chen YH, Carducci MA, Liu G, Jarrard DF, Hahn NM, et al. Chemohormonal therapy in metastatic hormone-sensitive prostate cancer: long-term survival analysis of the randomized phase, respectively III E3805 CHAARTED trial. J Clin Oncol 2018;36(11):1080–7. https://doi.org/10.1200/JCO.2017.75.3657.

[17] Tripathi A, Chen Y-H, Jarrard DF, Hahn NM, Garcia JA, Dreicer R, et al. Eight-year survival rates by baseline prognostic groups in patients with metastatic hormone-sensitive prostate cancer (mHSPC): an analysis from the ECOG-ACRIN 3805 (CHAARTED) trial. J Clin Oncol 2022;40(Suppl.16):5081. https://doi.org/10.1200/JCO.2022.40.16_suppl.5081.

[18] Vale CL, Fisher D, Godolphin P, Rydzewska LH, Boher J-M, Burdett S, et al. Defining more precisely the effects of docetaxel plus ADT for men with mHSPC: Meta-analysis of individual participant data from randomized trials. J Clin Oncol 2022;40(Suppl.16):5070. https://doi.org/10.1200/JCO.2022.40.16_suppl.5070.

[19] Vale CL, Fisher DJ, Godolphin PJ, Rydzewska LH, Boher JM, Burdett S, et al. Which patients with metastatic hormone-sensitive prostate cancer benefit from docetaxel: a systematic review and meta-analysis of individual participant data from randomised trials. Lancet Oncol 2023;24(7):783–97. https://doi.org/10.1016/S1470-2045(23)00230-9.

[20] Hussain M, Tombal B, Saad F, Fizazi K, Sternberg CN, Crawford ED, et al. Darolutamide plus androgen-deprivation therapy and docetaxel in metastatic hormone-sensitive prostate cancer by disease volume and risk subgroups in the phase III ARASENS trial. J Clin Oncol 2023:JCO2300041https://doi.org/10.1200/JCO.23.00041.

[21] Davis ID, Martin AJ, Zielinski RR, Thomson A, Tan TH, Sandhu S, et al. Updated overall survival outcomes in ENZAMET (ANZUP 1304), an international, cooperative group trial of enzalutamide in metastatic hormone-sensitive prostate cancer (mHSPC). J Clin Oncol 2022;40(Suppl.17):LBA5004. https://doi.org/10.1200/JCO.2022.40.17_suppl.LBA5004.

[22] Chi KN, Agarwal N, Bjartell A, Chung BH, Pereira de Santana Gomes AJ, Given R, et al. Apalutamide for metastatic, castration-sensitive prostate cancer. N Engl J Med 2019;381(1):13–24. https://doi.org/10.1056/NEJMoa1903307.

[23] Chi KN, Chowdhury S, Bjartell A, Chung BH, Pereira de Santana Gomes AJ, Given R, et al. Apalutamide in patients with metastatic castration-sensitive prostate cancer: final survival analysis of the randomized, double-blind, phase III TITAN study. J Clin Oncol 2021;39(20):2294–303. https://doi.org/10.1200/JCO.20.03488.

[24] Bjartell A, Agarwal N, Karsh L, Merseburger AS, Pereira de Santana Gomes AJ, Juárez Soto A, et al. Relationships of sites and burden of metastases with long-term outcomes and molecular subtypes in TITAN. Abstr MP24-08. J Urol 2021;206(Suppl.3):e414–5. https://doi.org/10.1097/JU.0000000000002015.08.

[25] Bjartell A, Ye D, Agarwal N, Chung BH, Given R, Merseburger A, et al. Apalutamide (APA) for metastatic castration-sensitive prostate cancer (mCSPC) in TITAN: outcomes in patients (pts) with de novo (D1) mCSPC vs. progression to mCSPC after localized disease (D0) at diagnosis. Eur Urol Open Sci 2020;19(Suppl.2):e863https://doi.org/10.1016/S2666-1683(20)33159-1.

[26] Chowdhury S, Bjartell A, Merseburger AS, Chung BH, Uemura H, Ye D, et al. Apalutamide (APA) for metastatic castration-sensitive prostate cancer (mCSPC): outcomes in high-volume (HV) and low-volume (LV) disease from the titan final analysis (FA). Eur Urol 2021;79(Suppl.1):S1180–1. https://doi.org/10.1016/S0302-2838(21)01220-3.

[27] Scher HI, Halabi S, Tannock I, Morris M, Sternberg CN, Carducci MA, et al. Design and end points of clinical trials for patients with progressive prostate cancer and castrate levels of testosterone: recommendations of the prostate cancer clinical trials working group. J Clin Oncol 2008;26(7):1148–59. https://doi.org/10.1200/JCO.2007.12.4487.

[28] Chi KN, Saad F, Chowdhury S, Graff JN, Agarwal N, Oudard S, et al. Prostate-specific antigen (PSA) kinetics in patients (pts) with advanced prostate cancer treated with apalutamide: results from the TITAN and SPARTAN studies. J Clin Oncol 2020;38(Suppl.15):5541. https://doi.org/10.1200/JCO.2020.38.15_suppl.5541.

[29] Chowdhury C, Bjartell A, Agarwal N, Chung BH, Given RW, Pereira de Santana Gomes AJ, et al. Apalutamide for metastatic castration-sensitive prostate cancer in TITAN: prognostic importance of prostate-specific antigen responses. J Urol 2020;203(4S):e250https://doi.org/10.1097/JU.0000000000000844.011.

[30] Chowdhury S, Bjartell A, Agarwal N, Chung BH, Given RW, Pereira de Santana Gomes AJ, et al. Deep, rapid, and durable prostate-specific antigen decline with apalutamide plus androgen deprivation therapy is associated with longer survival and improved clinical outcomes in TITAN patients with metastatic castration-sensitive prostate cancer. (ePub ahead of print]). Ann Oncol2023. https://doi.org/10.1016/j.annonc.2023.02.009.

UNCORRECTED PROOF

ARTICLE IN PRESS



[31] Shore ND, Armstrong AJ, Szmulewitz RZ, Petrylak DP, Holzbeierlein J, Villers A, et al. Efficacy of enzalutamide (ENZA) + androgen deprivation therapy (ADT) in metastatic hormone-sensitive prostate cancer (mHSPC) by pattern of metastatic spread: ARCHES post hoc analyses. J Clin Oncol 2020;38(Suppl.15):5547. https://doi.org/10.1200/JCO.2020.38.15_suppl.5547.

[32] Davis ID, Martin AJ, Stockler MR, Begbie S, Chi KN, Chowdhury S, et al. Enzalutamide with standard first-line therapy in metastatic prostate cancer. N Engl J Med 2019;381(2):121–31. https://doi.org/10.1056/NEJMoa1903835.

[33] National Comprehensive Cancer Network . Prostate cancer. Version 1. 2023-September 16; 2022. Available from: ⟨https://www.nccn.org/professionals/physician_gls/pdf/prostate.pdf⟩. [Accessed 15th November 2022].

[34] Smith MR, Hussain M, Saad F, Fizazi K, Sternberg CN, Crawford ED, et al. Darolutamide and survival in metastatic, hormone-sensitive prostate cancer. N Engl J Med 2022;386(12):1132–42. https://doi.org/10.1056/NEJMoa2119115.

[35] Chi KN, Merseburger AS, Ozguroglu M, Chowdhury S, Bjartell A, Chung B, et al. The effect of prior docetaxel (DOC) treatment on efficacy and safety of apalutamide (APA) plus androgen deprivation therapy (ADT) in patients (pts) with metastatic castration-sensitive prostate cancer (mCSPC) from TITAN. J Clin Oncol 2022;40(Suppl.6):89. https://doi.org/10.1200/JCO.2022.40.6_suppl.089.

UNCORRECTED PROOF